In this field, none of the d orbitals point dirctly towards the ligands,however the t_2 orbitals (d_{xy}, d_{yz} and d_{zx}) are pointing close to the direction in which ligands are approaching than the e orbitals ($d_{x^2-y^2}$ and d_{z^2}).

As a result, the energy of t_2 orbitals increases by $2/5\Delta_t$ and that of e orbitals decreases by $3/5\Delta_t$ as shown below. when compared to the octahedral field, this splitting is inverted and the spliting energy is less. The relation between the crystal field splitting energy in octahedral and tetrahedral ligand field is given by the expression; $\Delta_t = \frac{4}{9}\Delta_0$

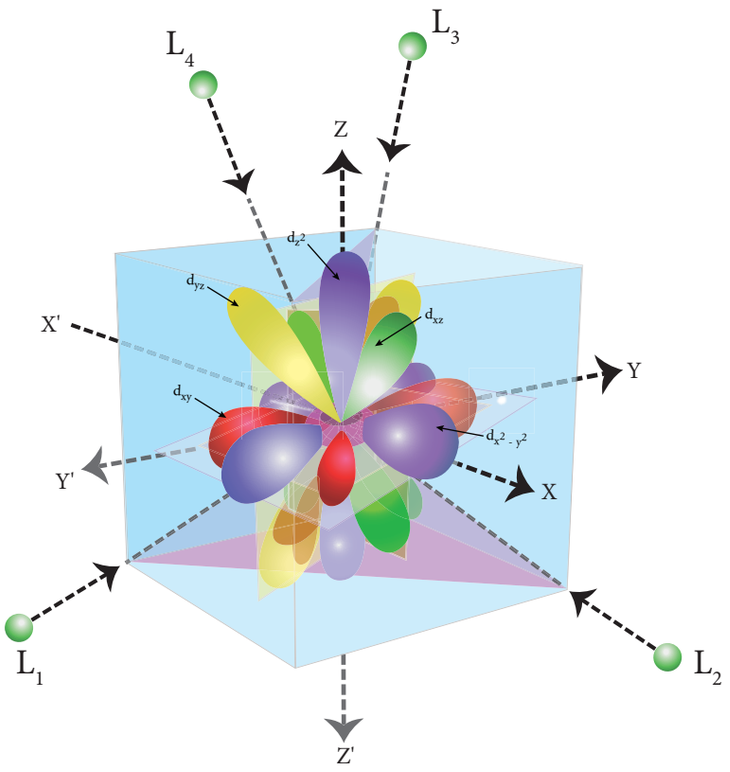


Figure 5.12 d-orbitals in tetrahedral ligand field

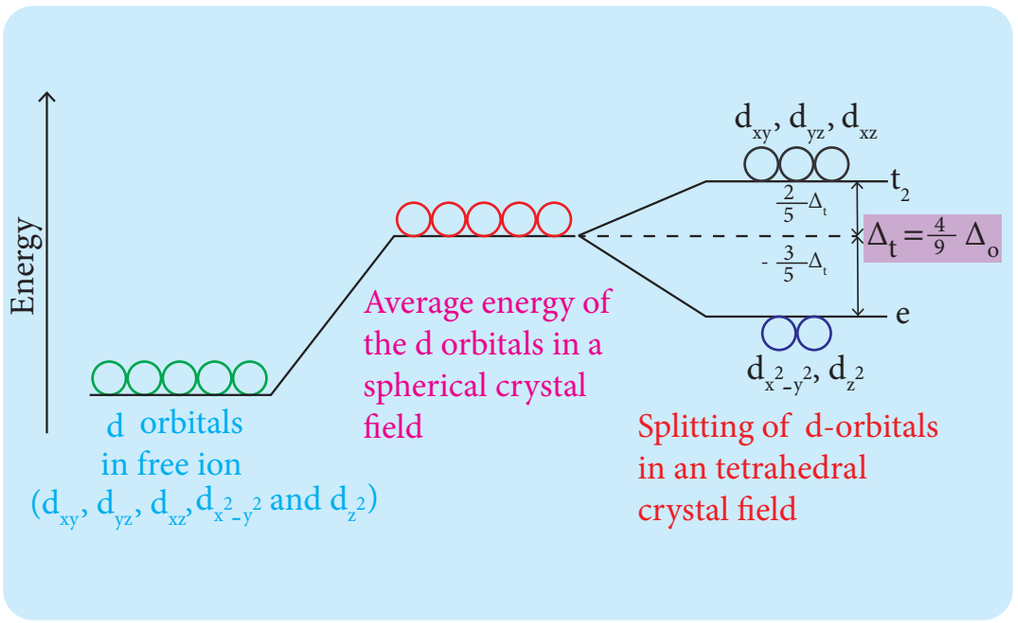


Figure: 5.13 - Crystal field splitting in tetrahedral field

Crystal filed splitting Energy and nature of ligands:

The magnitude of crystal field splitting energy not only depends on the ligand field as discussed above but also depends on the nature of the ligand, the nature of the central metal atom/ion and the charge on it. Let us understand the effect of the nature of ligand on crystal field splitting by calculating the crystal field splitting energy of the octahedral complexes of titanium(III) with different ligands such as fluoride, bromide and water

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